

BA890 - Research Project
Fake and Actual News Detection Model
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Motivation & Problem Statement:

The rise of fake news has become a significant issue with the proliferation of social media and online platforms, where information spreads rapidly without thorough fact-checking. Fake news can be intentionally deceptive, misleading, and can have serious consequences, such as influencing public opinion, spreading misinformation, and even impacting elections or public health. The ease with which false information can be shared amplifies its reach and impact, making it a pressing concern in today's digital age.

Developing a model to detect and classify fake news is crucial for combating misinformation and maintaining public trust in information sources. Such models can help prevent the spread of false information, protect democratic processes, and safeguard public health by ensuring people receive accurate and reliable news. Additionally, these models support journalistic integrity by flagging potentially fake news before it is disseminated, ultimately helping to preserve the integrity of information in society.

Dataset & Data Source:

The dataset comprises of new articles, fake and actual news pieces divided in two data frames each with 4 columns.

Dataset separated in two files:

1. Fake.csv (23502 fake news article)
2. True.csv (21417 true news article)

Dataset columns:

1. Title: title of news article
2. Text: body text of news article
3. Subject: subject of news article
4. Date: publish date of news article

Data Source and Download Link: [Download Link](#)

Research Methodology:

1. Data Cleaning and Pre-processing: The initial step involved cleaning the dataset by removing duplicates and handling missing values. Dates were standardized to a uniform format, and rows with errors (such as entries containing only URLs without any valid text) were dropped. Labels were assigned, with 0 indicating actual news and 1 indicating fake news. After this, initial visualizations were performed to understand the distribution and characteristics of the cleaned data. Finally, the fake and real news datasets were merged for further analysis.

2. Text Processing: Text processing is a critical step to prepare the data for machine learning models. We observed that the real news dataset often included the source and location at the beginning of articles (e.g., "ANKARA (Reuters)"). To ensure the model did not learn from these features, we removed the source and location information. Similarly, many entries in the fake news dataset began with "VIDEO," which was also removed. Further text preprocessing included converting all text to lowercase, removing non-alphanumeric characters, eliminating stop words, and tokenizing the text. Rows that were empty after cleaning (e.g., those containing only HTML links) were also removed.

3. Sentiment Analysis: To gain deeper insights into the dataset, sentiment analysis was conducted using the SentimentIntensityAnalyzer. It was hypothesized that fake news might exhibit more negative sentiment to attract and influence the audience, potentially serving as clickbait. The analysis revealed that the most common sentiment for actual news text was neutral, while fake news tended to be more negative. This trend was consistent across both the text and title columns, confirming the initial hypothesis.

4. Sentence Embeddings: To preserve context and meaning in the text, sentence embeddings were generated using the Huggingface SentenceTransformer('all-MiniLM-L6-v2') model. Separate embeddings were created for the title and text columns, allowing the classification models to learn from these features individually.

5. Classification Models: Four machine learning models were used for classification: Support Vector Machine (SVM), Decision Tree, Random Forest, and XGBoost. The models were trained and tested separately on the text and title columns. The best performance was observed with SVM, achieving an accuracy of 96.87% on the text and 94.53% on the title. Other models, such as Random Forest and XGBoost, also performed well, though the Decision Tree model showed comparatively lower accuracy.

Model	Text Test Accuracy	Text Training Accuracy	Title Test Accuracy	Title Training Accuracy
SVM	96.873274	96.471514	94.530773	93.669250
Decision Tree	84.333247	82.437075	80.802537	78.755236
Random Forest	93.034816	92.319741	90.485819	89.385410
XGBoost	95.510358	94.716894	93.146877	91.988302

6. Cosine Similarity Analysis: To assess whether titles and text were similar in context, potentially indicating clickbait, cosine similarity between the text and title was calculated. Contrary to expectations, the mean cosine similarity was similar for both fake and actual news, suggesting that titles in fake news were not drastically different from the text content. The threshold-based classification approach using cosine similarity did not yield a significant accuracy improvement.

Analysis Results:

The data cleaning and pre-processing steps successfully prepared the dataset for analysis, ensuring that the text used for model training was free from irrelevant information like source locations and common introductory phrases. Sentiment analysis provided valuable insights into the nature of

fake and real news, with fake news showing a tendency toward more negative sentiment, supporting the hypothesis about its potentially manipulative intent.

The classification models, particularly SVM and XGBoost, achieved high accuracy rates, indicating that the sentence embeddings effectively captured the distinguishing features between fake and real news. However, the cosine similarity analysis revealed that both fake and actual news titles were similar to their respective text content, implying that titles in fake news were not significantly misleading.

Overall, the methodology demonstrates a comprehensive approach to detecting and classifying fake news, leveraging advanced text processing techniques and machine learning models to achieve high accuracy in predictions.

Future Steps:

Moving forward, several enhancements could be made to improve the robustness and applicability of the fake news detection model. **Topic modeling** techniques, such as Latent Dirichlet Allocation (LDA), could be employed to uncover underlying themes and topics within the news articles, potentially revealing additional patterns that distinguish fake news from real news. Expanding the dataset to include **diverse sources** and languages would further enhance the model's generalizability, making it more effective across different contexts and regions. Additionally, incorporating **temporal analysis** could provide insights into how fake news trends evolve over time, enabling the model to adapt to changing patterns and emerging tactics. Finally, **real-world testing** of the model in live environments would be essential to evaluate its practical performance and effectiveness, providing valuable feedback to refine and improve the model's accuracy and usability in real-world applications.

Conclusion and Relevance:

The application of advanced machine learning techniques in natural language processing, as demonstrated by this project, makes a significant contribution to the broader field of AI-driven text analysis. This project is highly relevant in addressing critical issues of the digital age. By accurately detecting fake news, the model plays a crucial role in **combating misinformation**, thereby safeguarding public trust in media and information sources. This is vital for maintaining the integrity of democratic processes and public health, as reliable information is essential for informed decision-making. Once refined, integrating this model into social media platforms could significantly aid in **preventing and flagging fake news** before it spreads extensively. Additionally, the project raises awareness about the nature of fake news and the importance of information verification, which enhances **public awareness** and encourages more discerning consumption of news, ultimately fostering a more informed and responsible society.

Python Notebook on GitHub: <https://github.com/Parita2442/Fake-News-Detection>

Appendix

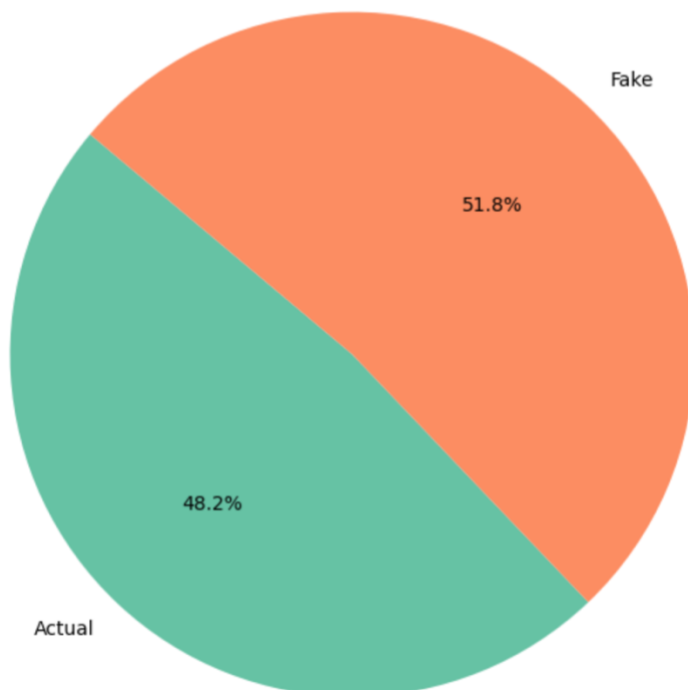
Word Cloud for Fake news title:

Most common word VIDEO

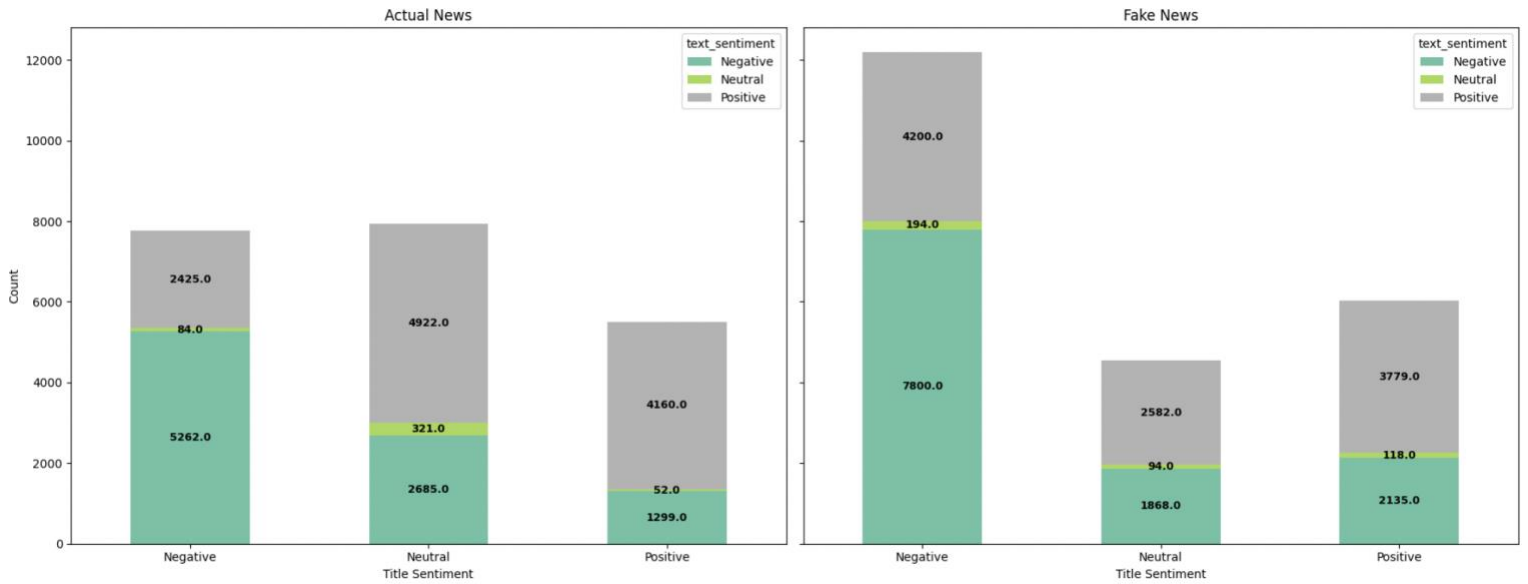


Dataset Distribution:

Distribution of Fake and Actual News



Sentiment Analysis:



Cosine Similarity Frequency distributions:

